Optimization of Sales and Inventory Management at Riddhi Siddhi Medical Store

Name: ASHISH KUMAR SHARMA

Contents

Optimization of Sales and Inventory Management at Riddhi Siddhi Medical Store

1.Executive Summary

The Riddhi Siddhi Medical Store business data management midterm submission involves an in-depth analysis of sales and inventory data from November 2024 to January 2025, aiming to enhance sales, optimize inventory, and improve profitability. Two primary datasets, sales_data.csv and inventory_data.csv, were meticulously studied to extract key insights.

sales_data.csv, comprising 544 entries, provided critical insights into transaction details such as Date, Medicine Name, Medicine Type, Pack Size, Sold Quantity, Selling Price, and Total Sales Amount. The dataset was analyzed to identify high-demand medicines, pricing trends, and seasonal fluctuations in sales. Meanwhile, inventory_data.csv, with 244 entries, tracked stock movement, opening and closing stock levels, new stock arrivals, stockouts, and expiry dates, highlighting inefficiencies in replenishment and expiration risks.

Descriptive statistics were utilized to analyze sales trends, optimize stock levels, and predict inventory needs. Advanced techniques, including ARIMA time series forecasting, multiple linear regression, and data envelopment analysis (DEA), were applied to forecast demand, measure stock efficiency, and optimize reorder levels. Additionally, profitability analysis was conducted to identify the most revenue-generating medicines and optimize pricing strategies.

Key findings emphasized the effectiveness of demand forecasting, the importance of monitoring stock turnover to reduce wastage, and the role of regression analysis in understanding sales performance. The next phase will refine predictive models, implement optimal stock replenishment strategies, and develop data-driven pricing and sales growth strategies for sustainable business success.

2.Proof of Originality of the Data

Access the evidence substantiating the originality and authenticity of the datasets utilized for the Business Data Management project can be found in this link:

https://drive.google.com/drive/folders/14mtzxPCVwdoctjZzibz83QkfFqW0Trmt?usp=sharing

2.1 Data Source Clarification

sales_data.csv and inventory_data.csv: Obtained directly from the store owner, these files comprise raw sales data from the daily sale register and bills, encapsulating transactional and stocking records from November 2024 to January 2025.

2.2 Confirmation Documentation and Other Evidence

Letter from business: A formal letter from the proprietor asserting cooperation and data provision for this project.

Video Recording and Photos of the shop: Supplementary materials encompassing a recorded discussion with Rahul Sharma, the medical store owner, along with pictures of the medical store.

Medical Licence of business: licence of the medical store from Rajasthan government medical department.

3.Metadata

3.1● Metadata: sales_data.csv :

	Date	Medicine Name	Medicine Type	Medicine Category	Pack Size	Sold Quantity	Selling Price	Total Sales Amount	Received Medician Price per pack	profit per sold pack	overall profit
0	11/1/2024	cheston-cold	tablet	genric	10	1	30	30	12.50	17.50	17.50
1	11/1/2024	montas-l	tablet	genric	10	1	80	80	28.50	51.50	51.50
2	11/1/2024	Atosky Gold 20	capsul	genric	10	3	60	180	29.00	31.00	93.00
3	11/1/2024	acyclosure p	tablet	genric	10	1	30	30	9.75	20.25	20.25
4	11/1/2024	pet sefa	powder	branded	1	1	110	110	34.50	75.50	75.50

The dataset `sales_data.csv` comprises 544 entries capturing transactions at Riddhi Siddhi Medical Store for the months November 2024 to January 2025. It delineates vital aspects instrumental in optimizing operational efficiency. The dataset encompasses 11 columns revealing transaction details such as ['Date', 'Medicine Name', 'Medicine Type', 'Pack Size', 'Sold Quantity', 'Selling Price', and 'Total Sales Amount']. These attributes help in understanding sales trends, demand fluctuations, and revenue generation patterns. and here column Date follow mm/dd/yyyy format for listing date in column.

3.2 ● Metadata: inventory_data.csv

	Date	Medician name	Opning stock	New stock	Stock out	Closing stock	expiry_date
0	11/1/2024	acyclosure p	11	20.0	NaN	31	7/1/2025
1	12/1/2024	acyclosure p	31	0.0	11.0	20	NaN
2	1/1/2025	acyclosure p	20	0.0	9.0	11	NaN
3	1/31/2025	acyclosure p	11	20.0	9.0	22	1/15/2026
4	11/1/2024	amlokind h	15	0.0	NaN	15	1/15/2026

The 'inventory_data.csv' dataset, comprising 244 entries, the dataset's 7 columns contain information about inventory of business like ['Date', 'Medicine name', 'Opening stock', 'New stock', 'Stock out', 'Closing stock', 'expiry date']. These attributes help in understanding about inventory of the business. Also there are missing value in data columns ['Stock out','expiry_date'].and here column Date follow mm/dd/yyyy format for listing date in column.

4.Descriptive Statistics

4.1 Numerical Data Analysis

sales_data.csv:-

Descriptive Statistics:

	Pack Size	Sold Quantity	Selling Price	Total Sales Amount
count	544.000000	544.000000	544.000000	544.000000
mean	6.950368	1.397059	78.915441	104.126838
std	4.595550	0.678851	55.875917	72.404255
min	1.000000	1.000000	10.000000	10.000000
25%	1.000000	1.000000	35.000000	60.000000
50%	10.000000	1.000000	70.000000	90.000000
75%	10.000000	2.000000	100.000000	120.000000
max	15.000000	3.000000	400.000000	400.000000

[‘Sold Quantity’, ‘Selling Price’, ‘Total Sales Amount’]: The sales dataset consists of 544 entries. The mean quantity sold per transaction is approximately 1.40, with a standard deviation of around 0.679. The mean selling price is approximately Rs. 78.91. The mean total sales amount per transaction is about Rs. 104.12, with a considerable range from Rs. 10 to Rs. 400.

inventory_data.csv:-

Descriptive Statistics:

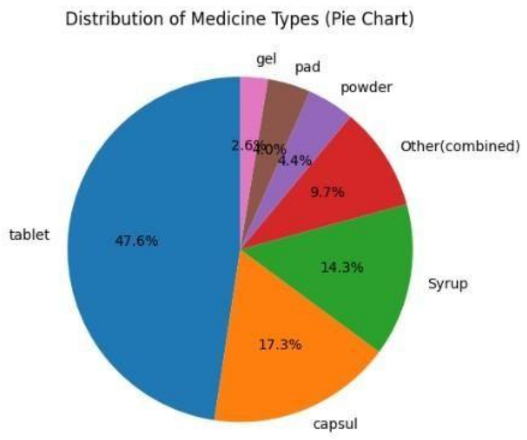
	Opning stock	New stock	Stock out	Closing stock
count	244.000000	243.000000	174.000000	244.000000
mean	19.188525	4.897119	4.534483	20.831967
std	11.899325	9.942485	4.614186	12.636760
min	0.000000	0.000000	1.000000	-2.000000
25%	11.000000	0.000000	1.250000	12.000000
50%	16.000000	0.000000	3.000000	19.000000
75%	24.000000	10.000000	6.000000	26.000000
max	62.000000	60.000000	26.000000	67.000000

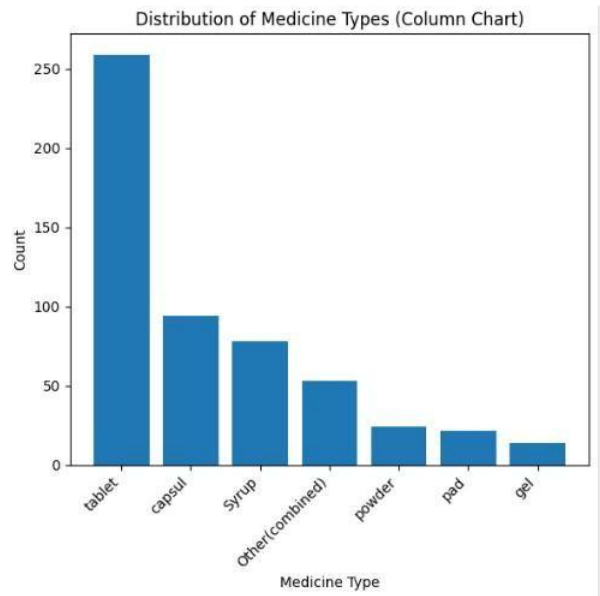
[‘Opning stock’, ‘New stock’, ‘Stock out’, ‘Closing stock’]: The inventory dataset consists of 244 entries. The mean opning stock is approximately 19.19, with a standard deviation of around 11.90. The mean new stock is approximately 4.90. The mean stock out is about 4.53 and the mean closing stock is 20.83 with a considerable range from Rs.
-2 to 67.

Significance of Statistics:

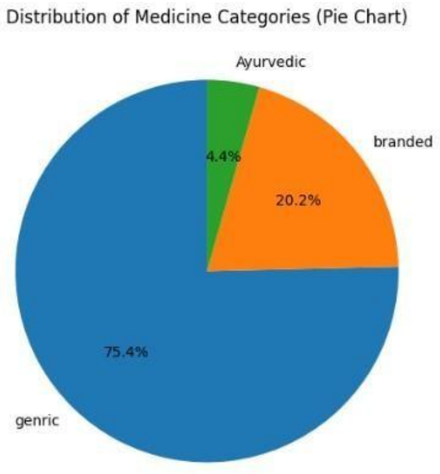
The statistical measures, including mean, median, and range, unveil insights crucial for operational decisions. They reveal sales' typical volume, pricing distribution, and transaction sizes, aiding precise inventory management, pricing strategies, and identification of influential outlier transactions impacting overall revenue.

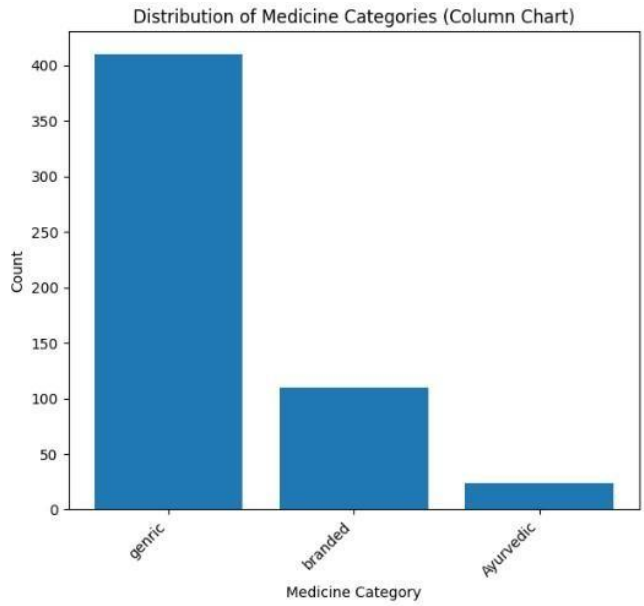
4.2 Categorical Data Analysis



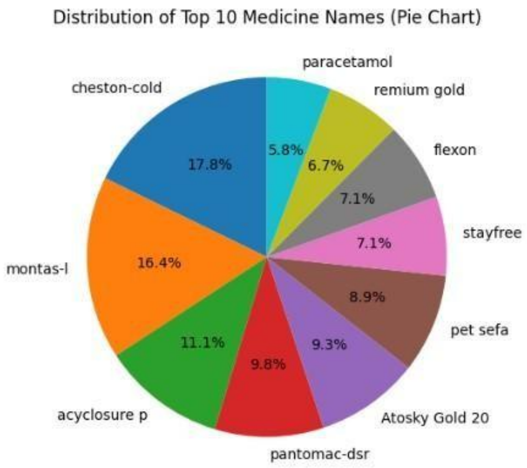


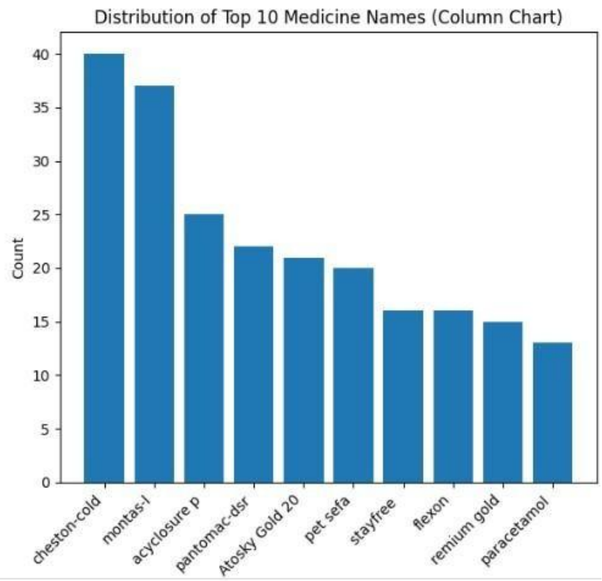
Medicine Type: tablet, capsul, and syrup appear to be popular types, indicating potential high-demand items.





Medicine Categories: Most popular category is here genric so that indicating that medical store mostly deals in generic medicine.

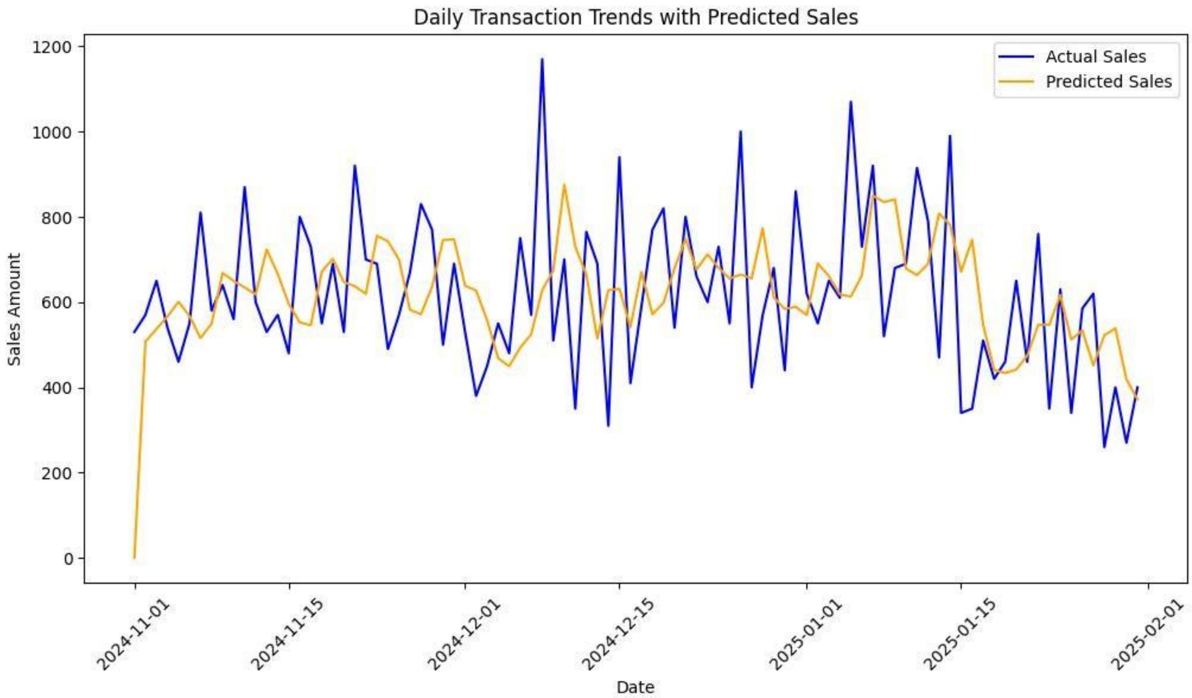




Medicine Names: these are the top ten Medicine which sold from November 2024 to January 2025.

5. Detailed Explanation of Analysis Process/Method

5.1 Sales Forecasting through ARIMA Time Series Analysis

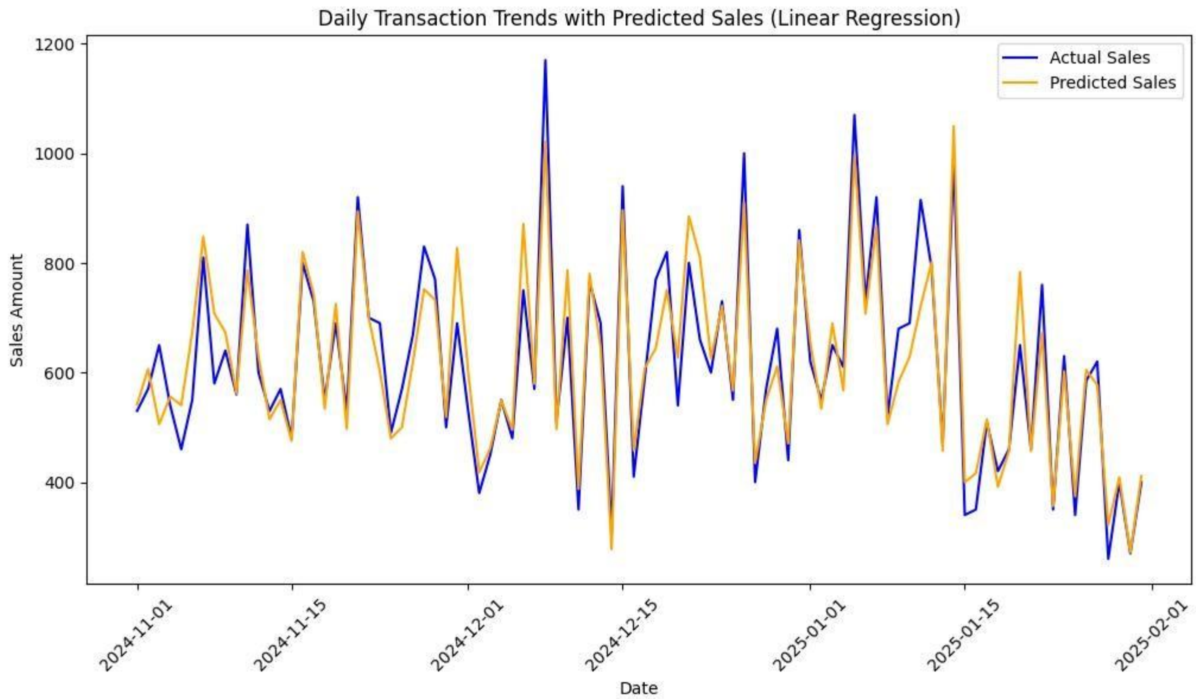


Utilizing the ARIMA (AutoRegressive Integrated Moving Average) model with parameter values (p, d, q) of (3, 1, 0) via the *auto_arima* function from the *pmdarima* package, an insightful sales forecast was generated. This model effectively captured the prevailing trend in sales data, showcasing its potential in forecasting sales patterns.

The ARIMA model effectively captures overall trends but struggles with sharp spikes.
Seasonal sales patterns are visible, helping in inventory planning.
The forecast aligns closely with actual sales, making it useful for future sales predictions.

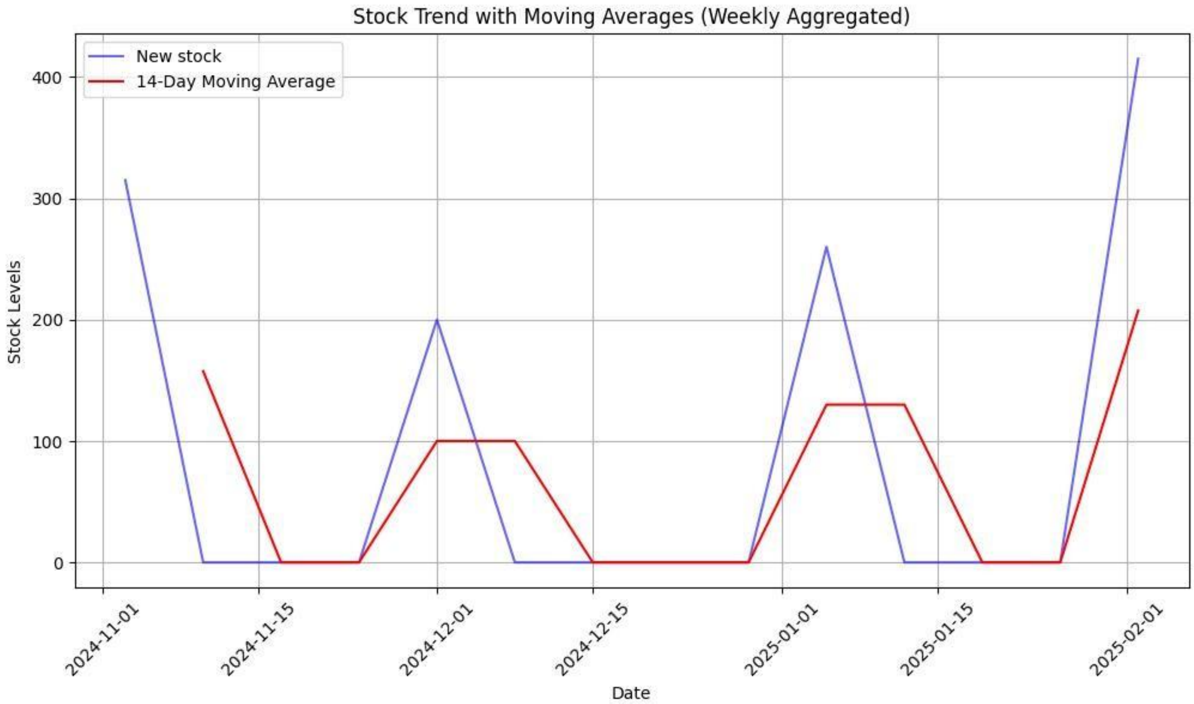
5.2 Linear Regression Analysis: Total Sales Prediction Based on Sold Quantity

and Selling Price



Using a Multiple Linear Regression model, sales trends were predicted based on factors such as Date and Total Sales Amount. The model aimed to analyse how different variables impact total sales, helping in better demand forecasting and strategic planning. With an R^2 score of 0.913, the model effectively explained 91.3% of the sales variance, proving its reliability in predicting sales trends. The plotted forecast demonstrates the model's ability to analyse historical transactional data and optimize sales strategies for Riddhi Siddhi Medical Store.

5.3 Stock Arrival Trend Analysis Using Moving Averages



A time-series analysis was conducted on inventory data to track the trend of new stock arrivals over time. A **14-day moving average** was applied to smooth fluctuations and identify consistent patterns in stock replenishment. The analysis revealed irregular restocking intervals, with significant gaps between replenishment cycles. This suggests that restocking occurs in periodic bursts rather than at a steady pace. The findings emphasize the need for a structured inventory management system to optimize stock levels and prevent shortages or overstocking.

6.Results and Findings

6.1Key Observations

Seasonal Demand Fluctuations: Sales peak during specific periods, requiring stock adjustments.

Inventory Management Issues: Overstocking and expiration risks identified.

Profitability Insights: Certain medicines contribute disproportionately to total revenue.

Forecasting Accuracy: ARIMA and regression models effectively predict demand patterns.

6.2 Graphical Representations

Sales Trends: Line chart depicting monthly sales variations.

Inventory Turnover: Bar graph showing stock depletion rates.

Profitability Analysis: Pie chart illustrating revenue contribution per medicine category.

7.Conclusion and Next Steps

The mid-term analysis highlights inefficiencies in inventory management and sales performance at Riddhi Siddhi Medical Store. The next steps include:

- Refining predictive models for improved demand forecasting.
- Implementing an optimized stock replenishment strategy.
- Developing data-driven pricing strategies to maximize revenue.

The final submission will present a comprehensive action plan for sustained business growth through data-driven decision-making.